

Executive Summary

The **Customer Behavior Analysis** project was developed to analyze customer purchasing patterns, identify customer segments, and study churn behavior using machine learning and data analytics techniques. The dataset includes customer transaction details, product information, payment methods, demographic data, and churn status.

The project involved data cleaning, preprocessing, exploratory data analysis (EDA), feature engineering, RFM analysis, and customer segmentation using the K-Means clustering algorithm. Various visualizations such as bar charts, line graphs, scatter plots, histograms, and heatmaps were used to understand customer behavior and purchasing trends.

The analysis helped identify high-value customers, churn-risk customers, and top-performing product categories. Based on the findings, business recommendations such as loyalty programs, personalized marketing, and customer retention strategies were suggested for Alfido Tech.

This project demonstrates how data analytics and machine learning can help businesses make data-driven decisions, improve customer engagement, and increase overall business performance.

Code Links

https://github.com/Shivani4002/Alfido-Tech-Internship/blob/main/Customer_Behavior_Analysis/Customer_Behavior_Analysis.ipynb

Screenshot & Visualization

